

Replicating Deep Reinforcement Learning for Stock Trading: A Python Study with Nested Walk-Forward Tuning

IFTE0004 Personal Coursework

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Selected Paper: Yang et al. (2020) — Deep Reinforcement Learning for Automated Stock Trading. ICAIF 2020.

Github link: <https://github.com/hoisaac484/yang-2020-drl-trading-replication>

1. Description of the Paper

Automated stock trading is among the most prominent tasks in quantitative finance, portfolio managers are supposed to select stocks to trade, their size, and timing while considering transaction costs, liquidity, and portfolio constraints. Traditionally, portfolio management models were divided into two sequential parts: return prediction and optimal allocation, accumulating inaccuracies at both levels and making adaptation challenging. Yang et al. (2020) attempt to overcome this problem by formulating trading as a sequential portfolio control task using deep reinforcement learning (DRL): the agent directly observes market state variables and chooses a continuous trade size, integrating transaction costs and portfolio constraints within one optimisation target.

For this purpose, the agent observes a 181-dimensional state space, including cash balance, adjusted closing prices, shares held, and four technical indicators (MACD, RSI, CCI, ADX) for each of 30 Dow companies. Actions represent a continuous vector from $[-1, 1]$ for each stock, scaled by a maximum-shares coefficient. The reward function was defined as the net increase in portfolio value minus 0.1% of transaction costs. Negative shares were disallowed and negative cash could only appear due to liquidity problems. Additionally, a financial turbulence index forced liquidation whenever required.

To construct a robust trading strategy, the authors trained PPO (Schulman et al., 2017), A2C (Mnih et al., 2016), and DDPG (Lillicrap et al., 2019) algorithms every quarter on an increasing historical window. All of them were evaluated against the last three months and an algorithm with the best Sharpe ratio was selected for trading in the next quarter. It relies on the idea that different algorithms demonstrate better results under various market conditions.

During 2016-2020, DRL-based trading on Dow 30 stocks provided by Compustat/WRDS demonstrated a cumulative return of 70.4% with annualised volatility of 9.7% and Sharpe ratio of 1.30 (Yang et al., 2020). Moreover, compared with DJIA (0.47) and minimum variance portfolio (0.45), this model yielded higher results and was protected by the proposed turbulence mechanism (-9.7%) while other strategies lost over 37% during March 2020 crash.

2. Python Implementation and Empirical Replication

The replication retains the paper's setting environment and logic while shifting the sample period to 2015-2026 with daily OHLCV data for the Dow 30 and DIA for benchmark proxies from yfinance. Stable-baselines3 (Raffin et al., 2021), rather than stable-baselines used by the paper, was selected due to having well-maintained, tested implementations of PPO, A2C, and DDPG with a consistent API. Thus, some internals of PPO/A2C/DDPG differ slightly from the paper.

The daily OHLCV is transformed into an adjusted-price panel. The technical indicator calculation uses only lagged information such that state at time t is independent of future prices. Return calculations for benchmark indices use the adjusted closing price. The data is processed strictly: if even one ticker has incomplete information for any required window, the pipeline fails, and data issues become apparent instead of silently performing forward-filling. The reward is rescaled by 10^{-4} , allowing the gradients' magnitudes during actor critic updates to stay numerically stable, otherwise, large dollar amount portfolios cause the gradient to be unstable.

The trading environment has 181-state (cash, prices, holdings, MACD, RSI, CCI, ADX per stock), continuous actions with $[-1, 1]$ range and scaled with the constant $H_{\max} = 100$, \$1,000,000 starting cash and 0.1% per trade transaction costs. The reward is defined as the change in portfolio value and is rescaled by 10^{-4} for stability. The turbulence threshold uses the adaptive author's repository version based on 90th percentile of the in-sample turbulence (Yang et al., 2020).

2.1 Experiment 1 — Author-Style Rolling Ensemble

The training is done in an expanding window, validation/trading window consists of 63 trading days, approximating the paper's quarterly cycle. Timesteps for A2C 30k, PPO 100k, and DDPG 10k are taken from the author's repository (Yang et al., 2020). Cash and positions are saved from previous windows to create continuous equity paths for 14 consecutive quarters since January 2023 onwards.

Rolling ensemble logic (Experiments 1 and 2):

```
for each 63-trading-day window:
    train PPO, A2C, DDPG on expanding history
    validate all three on latest validation window
    select agent with highest validation Sharpe
    trade next window; carry cash and holdings forward
```

2.2 Experiment 2 — Nested Walk-Forward Tuning Extension

There are three potential limitations justifying a follow-up study: a single 63-day validation window is too noisy, the authors' fixed hyperparameters might be suboptimal for the 2023-2026 period, and using a single seed will either inflate or deflate expected results. Experiment 2 evaluates candidate hyperparameters based solely on the previously performed validation folds over 5 different seeds and scores candidates according to $\text{mean}(\text{validation Sharpe}) - 0.5 \times \text{std}(\text{validation Sharpe})$ metric. Two possible validation window sizes and three different folds cap are tested (2 weeks, 4 weeks, and unlimited). The first candidate is trained using author's style since there is no prior validation fold available. The first fold size is set to 63 and then increased up to 126.

Nested tuning rule (Experiment 2):

```
for each trading window:
    collect validation folds strictly before the trading date
    for each candidate setting, across 5 seeds:
        score = mean(validation Sharpe) - 0.5 × std(validation Sharpe)
    select best-scoring candidate
    train PPO/A2C/DDPG; select trading agent by validation Sharpe
```

Trading universe is currently existing Dow 30, however, Amazon (AMZN), Nvidia (NVDA), and Sherwin Williams (SHW), and have been excluded because they weren't in the Dow index at all periods. Minimum-variance covariance matrix estimation is based on return data from the training period to avoid look-ahead bias.

3. Presentation of Results

These results have been selected due to their relevance to testing the core thesis of the paper: that of whether validation-Sharpe DRL ensemble is superior to passive and portfolio optimisation in risk-adjusted trading performance using an adjusted out-of-sample period. Table 1 compares replication results to those of the paper. Table 2 compares all strategies to each other in out-of-sample period. Results of Experiment 2 design sensitivity analysis can be found in Appendix B.

Table 1 compares replication results to those in the paper. Methodology remains intact but performance is far off, since Experiment 1 Sharpe (0.739) is about half the paper's (1.30) and drawdown is nearly twice as large, considering that the sample period, universe, and the library differ.

Study / Design	Period	Cum. Return	Sharpe	Max DD
Yang et al. ensemble	2016–2020	70.4%	1.30	−9.7%
Author-style replication	2023–2026	41.7%	0.739	−21.3%
Nested tuned extension	2023–2026	86.5%	1.254	−15.4%

Table 1. Original paper from Yang et al. (2020) versus replication.

Figure 1 and Table 2 displays the main out-of-sample results for years starting 2023 onwards. Nest-tuned RL outperforms DIA on both measures. However, equal-weight Dow 30 and min-var portfolio continue being extremely competitive, thus making the empirical finding a mixed case instead of promotional one.



Figure 1. Cumulative Returns Over Time: RL Strategies and Benchmarks, 2023 onward.

Strategy	Final Value	Cum. Return	Sharpe	Max DD
Author-style RL ensemble	\$1.417M	41.7%	0.739	−21.3%
Nested tuned RL (126 days, 5 seeds, folds=3)	\$1.865M	86.5%	1.254	−15.4%
DIA buy-and-hold	\$1.584M	58.4%	1.092	−16.0%
Equal-weight Dow 30	\$2.169M	116.9%	1.679	−19.1%
Minimum-variance	\$1.710M	71.0%	1.478	−9.9%

Table 2. Main out-of-sample results, 2023 onward.

Experiment 2 design sensitivity results (Table B1) show two separable effects. 126-day window is better in general compared to the 63/126-day window combination with the same fold cap. Moreover, fold cap effect is independent of window length, as it is demonstrated by the poor result of 126-day window and folds ≤ 1 (0.661 Sharpe) and the great result of 126-day window and folds ≤ 3 (1.254). Using only a recent single prior fold makes hyperparameters susceptible to recency bias in relation to one market regime. This effect is also shown by robustness check with respect to random seed used for folds (Table A1): Sharpe for Experiment 1 varies from 0.006 to 1.594 among different seeds.

4. Critical Comparison and Discussion

Replication of Yang et al.'s work confirms their key methodological result: the validation-Sharpe ensemble outperforms any individual agent in the study, and the framework is flexible enough, as the highest Sharpe (Experiment 2) is 1.254 compared with 1.30 in the original paper but based on a different market cycle. The completed Experiment 2 grid suggests another practical insight regarding the structure of the algorithm: it's critical to use the cap, whereas the choice of the length of the validation window matters less. The Sharpe achieved with a 126-day and folds ≤ 1 (0.661) configuration underperforms author-style Experiment 1 (0.739) despite having longer validation, as relying on a single recent prior fold introduces recency bias in choosing hyperparameters. Also, Table A1 indicates that the range of Sharpe for each experiment is large (0.006 to 1.594) regardless of the setup, thus, any single backtest is just one possible realisation, not an expected return.

There are four other reasons why Experiment 1 underperformed the results of the paper. First, 2023–2026 is a rising market characterised by several momentum mega-cap stocks, which implies that a passive equal-weight benchmark will have better returns. Second, daily technical indicator features are of limited usefulness in this environment. Third, shorter 63-day validation windows contribute to the noise in selecting models. Fourth, unstable DRL actions suffer from the transaction costs repeatedly charged at 0.1% per trade over the span of 14 quarters.

The bootstrap findings in Appendix C confirm the need for caution. With 10,000 samples, a 95% confidence interval for the Sharpe ratio is [0.06, 2.70] days. Comparing nested reinforcement learning with two simple baseline strategies via daily returns t-test shows no statistical advantage: against Dow Jones Industrial Average (DIA, $p = 0.303$) and against equal weight ($p = 0.446$). Given the large variance and non-significant p-value, it seems reasonable that any potential outperformance of reinforcement learning in Experiment 2 was driven by noise. A block bootstrap or Newey-West test would be preferable for the analysis in the presence of autocorrelations and heteroskedasticity among trading returns (Newey & West, 1987), although the independent and identically distributed resamples provided here are useful purely for descriptive purposes.

Both benchmarks are difficult to beat due to the financial reasoning that underlies this sample. On one hand, equal-weight strategy is advantageous due to broad exposure to rising 2023–2026 equities, particularly momentum mega-caps. On the other hand, the low variance portfolio benefits from diversification and low negative exposure. In order to justify its added complexity, RL needs to perform better in one or both areas – that is, better timing or risk management. In this particular case, the RL beats Dow Jones but doesn't convincingly outperform equally straightforward portfolios.

From the machine learning perspective, the main takeaway is that the selection of models also requires modelling. The risk of overfitting to a particular regime during the validation phase persists even with a sufficiently long trading window if the latter starts before the validation, so Experiment 2 solves the problem with a penalised scoring. From an econometrics perspective, a bootstrap shows that a high Sharpe ratio achieved in one backtest is very noisy even for a relatively stable strategy.

The most significant threat to external validity is the current universe of Dow 30 stocks: taking the same stock index backward in time to 2015 means incorporating the returns of AMZN, NVDA and SHW into the model – all companies with extremely high Sharpe ratios. This form of survivorship bias means that any RL advantage will be harder to attribute to true alpha. Other potential threats are summarised in Table 4.

Threat	Why it matters for interpretation
Fixed current Dow 30	Index-membership bias; AMZN/NVDA/SHW were not Dow members throughout 2015–2026.
Single seed (Experiment 1)	Five-seed check shows Sharpe ranging 0.006–1.594; the point estimate is one realisation.
Short 63-day validation	Noisy model selection; one unusual quarter can drive ensemble choice.
Small Experiment 2 grid	Tests validation-window and fold design only; broader hyperparameter search omitted.
No statistical significance	Bootstrap and paired tests do not confirm RL outperformance over DIA or equal-weight.

Table 4. Main threats to validity and their implications for interpretation.

The defensible conclusion is as follows: the Yang-style ensemble can indeed be replicated and extended, but the empirical evidence from the updated data is conditional. While the nested tuning procedure represents a methodological advance by demonstrating that longer validation windows and a penalised scoring system help to overcome regime-specific noise, none of experiments yield robust performance improvements compared to passive benchmarks. The framework offers an interesting way to approach portfolio construction from the angle of dynamic portfolio control – but the current results suggest that deep reinforcement learning is still not ready to compete with basic portfolio rules. One way to improve robustness is to test alternative levels of transaction cost from 0.05% to 0.20%.

References

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Appendix A. Experiment 1 Multi-Seed Robustness

The rolling ensemble was repeated across five random seeds to test sensitivity to stochastic initialisation and training. The wide dispersion confirms the fixed-seed result is one realisation.

Statistic	Final Value	Sharpe	Max DD
Mean	\$1.755M	1.038	−24.7%
Std	\$0.531M	0.643	10.6 pp
Min	\$1.005M	0.006	−43.4%
Max	\$2.468M	1.594	−18.1%

Table A1. Author-style five-seed dispersion (seeds 1–5).

Appendix B. Experiment 2 Nested Tuning Variants

All five nested tuning configurations, ranked by cumulative return. ‘Best’ means best among tested historical configurations only; it is not a guarantee of future optimality.

Experiment 2 configuration	Cum. Return	Sharpe	Max DD
126 days validation, 5 seeds, folds ≤ 3	86.5%	1.254	−15.4%
63/126 days validation, 5 seeds, folds ≤ 1	84.2%	1.094	−19.9%
63/126 days validation, 5 seeds, folds ≤ 3	66.3%	1.135	−11.1%
63/126 days validation, 5 seeds, all folds	60.6%	0.924	−20.2%
126 days validation, 5 seeds, folds ≤ 1	37.9%	0.661	−20.3%

Table B1. All nested tuning variants. Author-style Experiment 1 (Sharpe 0.739) shown for comparison.

Appendix C. Statistical Uncertainty Diagnostics

Bootstrap intervals use 10,000 independent and identically distributed daily-return resamples. Because daily returns are serially dependent, these intervals should be interpreted as descriptive uncertainty checks rather than formal inference.

Strategy	Sharpe	95% Bootstrap CI	Cum. Return CI
Nested tuned RL	1.254	[0.06, 2.70]	[3.3%, 236.4%]
Author-style RL	0.739	[−0.35, 2.04]	[−16.4%, 140.7%]
DIA	1.117	[−0.04, 2.44]	[−1.7%, 154.5%]
Equal-weight 30	1.730	[0.46, 3.20]	[25.7%, 269.0%]
Minimum-variance	1.505	[0.30, 2.82]	[12.1%, 155.1%]

Table C1. Bootstrap uncertainty for main performance metrics.

Comparison	Mean daily diff.	p-value	Sharpe diff. CI
Nested RL – DIA	0.02%	0.303	[−0.65, 0.99]
Nested RL – Equal-weight	−0.02%	0.446	[−1.29, 0.38]
Nested RL – Min-var	0.01%	0.656	[−1.48, 1.08]
Nested RL – Author-style	0.03%	0.183	[−0.40, 1.48]

Table C2. Paired daily return tests and bootstrap Sharpe-difference intervals.